

Week 12 LLM GA

1) Consider a problem of length extrapolation using an Absolute Position Encoding (APE) scheme. Suppose the context length of a model during training is 512. Which of the following approaches allows the model to extrapolate to a context length of 1024 tokens **1 point**

☒ Fixed Sinusoidal Encoding

☐ Parameterized APE

☐ ALiBi

☐ T5

512 → 1024

Yes, the answer is correct.

Score: 1

Accepted Answers:

Fixed Sinusoidal Encoding

2) The statement that, in principle, no APE schemes can encode the relative position information of tokens in the input sequence is **1 point**

☐ True

☒ False

Yes, the answer is correct

3) The pre-attention score e_{ij} in Relative Position Encoding (RPE) can be written in matrix form as follows **1 point**

$E = XW_Q(XW_K + P_K)^T$ What will be the dimension of P_K ? In all the options, T denotes context length.

☐ $T \times T \times T \times d_{model}$

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☐ $T \times T$

4) Suppose the context length of a model is 16K. Amongst the given choices, choose the RPE schemes that require the least number of parameters to encode the relative position information of the tokens. **1 point**

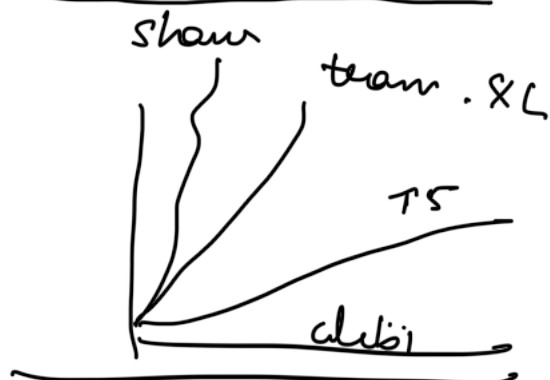
☐ T5-bias

☐ Transformer-XL

☐ Shaw RPE

☒ ALiBi

was modified to reach here



→ APE & RoPE

5) Select all the fixed position encoding schemes (i.e., it has no learnable components). Note that you may refer to the original papers if required. Avoid going through implementations of the papers in DL frameworks.

1 point

☐ T5-bias

☒ Transformer-XL

☒ RoPE

☒ ALiBi

No, the answer is incorrect.

Score: 0

Accepted Answers:

Transformer-XL

RoPE

ALiBi

6) Suppose that a team pre-trains a decoder model using CLM (Causal Language Modelling) objective using the APE scheme. However, during inference, one of the team members suggested not to add the position information for the following reasons

[I] The causal mask in the decoder itself encodes the position information as suggested in the NoPos paper

[II] Adding no additional positional embedding to each token will also help increase the inference speed marginally

☒ I is a correct reason but it can't be used only during inference as it hurts the performance

☒ II is a correct statement